Mabuhay Homes Community Portal — Complete ERD & DFD Audit Report

Comprehensive Architectural Analysis & Database Entity Audit

Audit Date: 2026-10-13 | Total Verified Database Entities: 9 | Technology Stack: Next.js 16.3 + MongoDB (Mongoose 9.9)

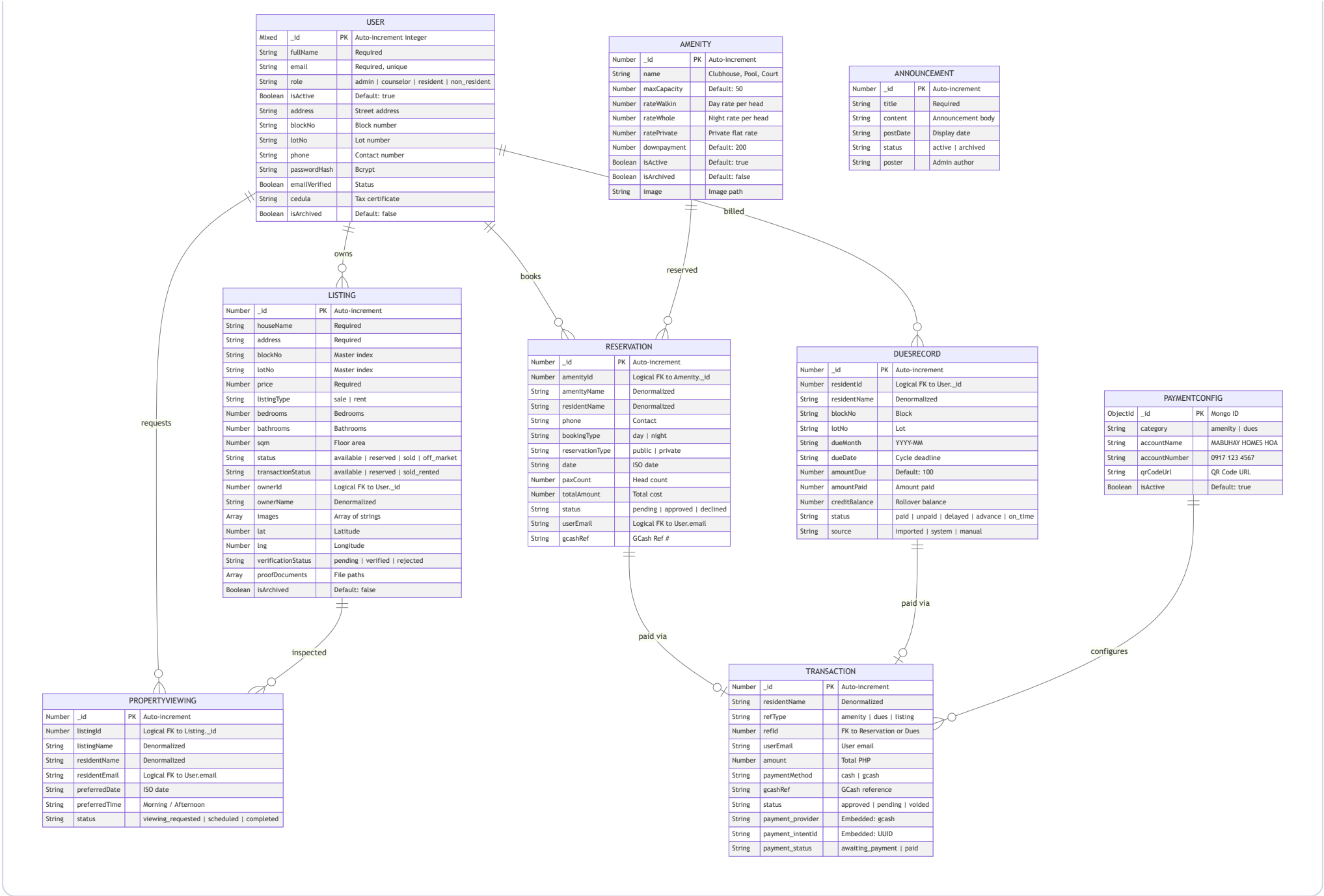
1. Executive Summary & Entity Audit Verdict

> **Audit Confirmation:** Ang inyong system ay may eksaktong **9 database entities (MongoDB collections)**. Ang paunang detection na 9 entities ay 100% tumpak. Hindi ito error o kulang sa scan — ito ay dahil sa NoSQL document embedding at denormalization na katangian ng MongoDB.

#	Entity / Mongoose Model	Collection Name	Primary Key	Core Purpose
1	**User**	`users`	_id (Number)	Resident, Admin, & Counselor accounts, auth hashes, tokens, block/lot.
2	**Listing**	`listings`	_id (Number)	Properties for sale/rent, ownership verification documents, geo-coordinates.
3	**Amenity**	`amenities`	_id (Number)	Clubhouse, basketball court, swimming pool, rates, capacity, rules.
4	**Reservation**	`reservations`	_id (Number)	Amenity bookings, date, guest counts, approval workflow.
5	**DuesRecord**	`duesrecords`	_id (Number)	Monthly HOA dues per resident, penalty computations, import tracking.
6	**Transaction**	`transactions`	_id (Number)	Financial ledger for reservations & dues, embedded GCash subdocument.
7	**PropertyViewing**	`propertyviewings`	_id (Number)	Resident/client requests to inspect listed properties.
8	**Announcement**	`announcements`	_id (Number)	Community bulletins, news, event notices, alerts.
9	**PaymentConfig**	`paymentconfigs`	_id (ObjectId)	HOA GCash account details, mobile number, active QR code.

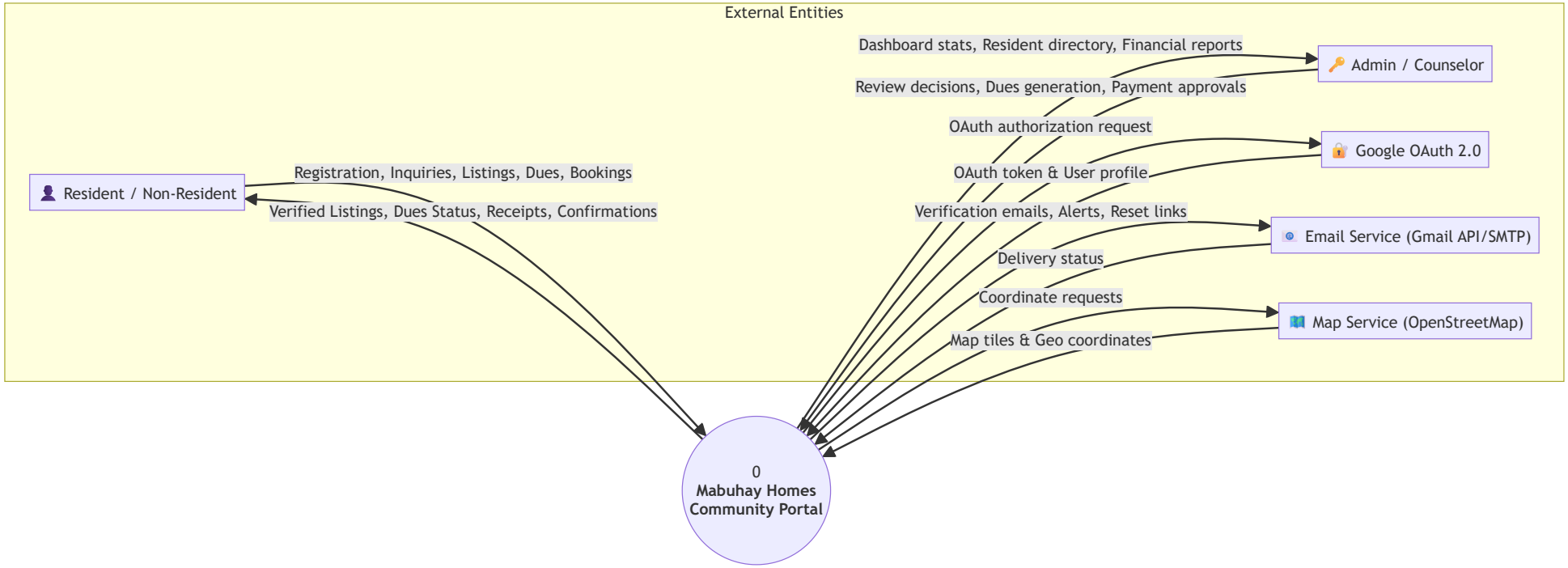
2. Complete Entity Relationship Diagram (ERD)

Displays all 9 verified MongoDB entities, their schema attributes, data types, primary keys, and logical relationships.


USER
Mixed _id PK Auto-increment integer
String fullName Required
String email Required, unique
String role admin | counselor | resident | non_resident
Boolean isActive Default: true
String address Street address
String blockNo Block number
String lotNo Lot number
String phone Contact number
String passwordHash Bcrypt
Boolean emailVerified Status
String cedula Tax certificate
Boolean isArchived Default: false
AMENITY
Number _id PK Auto-increment
String name Clubhouse, Pool, Court
Number maxCapacity Default: 50
Number rateWalkin Day rate per head
Number rateWhole Night rate per head
Number ratePrivate Private flat rate
Number downpayment Default: 200
Boolean isActive Default: true
Boolean isArchived Default: false
String image Image path
ANNOUNCEMENT
Number _id PK Auto-increment
String title Required
String content Announcement body
String postDate Display date
String status active | archived
String poster Admin author
owns
books
reserved
billed
requests
LISTING
Number _id PK Auto-increment
String houseName Required
String address Required
String blockNo Master index
String lotNo Master index
Number price Required
String listingType sale | rent
Number bedrooms Bedrooms
Number bathrooms Bathrooms
Number sqm Floor area
String status available | reserved | sold | off_market
String transactionStatus available | reserved | sold_rented
Number ownerId Logical FK to User._id
String ownerName Denormalized
Array images Array of strings
Number lat Latitude
Number lng Longitude
String verificationStatus pending | verified | rejected
Array proofDocuments File paths
Boolean isArchived Default: false
RESERVATION
Number _id PK Auto-increment
Number amenityId Logical FK to Amenity._id
String amenityName Denormalized
String residentName Denormalized
String phone Contact
String bookingType day | night
String reservationType public | private
String date ISO date
Number paxCount Head count
Number totalAmount Total cost
String status pending | approved | declined
String userEmail Logical FK to User.email
String gcashRef GCash Ref #
DUESRECORD
Number _id PK Auto-increment
Number residentId Logical FK to User._id
String residentName Denormalized
String blockNo Block
String lotNo Lot
String dueMonth YYYY-MM
String dueDate Cycle deadline
Number amountDue Default: 100
Number amountPaid Amount paid
Number creditBalance Rollover balance
String status paid | unpaid | delayed | advance | on_time
String source imported | system | manual
PAYMENTCONFIG
ObjectId _id PK Mongo ID
String category amenity | dues
String accountName MABUHAY HOMES HOA
String accountNumber 0917 123 4567
String qrCodeUrl QR Code URL
Boolean isActive Default: true
inspected
paid via
paid via
configures
PROPERTYVIEWING
Number _id PK Auto-increment
Number listingId Logical FK to Listing._id
String listingName Denormalized
String residentName Denormalized
String residentEmail Logical FK to User.email
String preferredDate ISO date
String preferredTime Morning / Afternoon
String status viewing_requested | scheduled | completed
TRANSACTION
Number _id PK Auto-increment
String residentName Denormalized
String refType amenity | dues | listing
Number refId FK to Reservation or Dues
String userEmail User email
Number amount Total PHP
String paymentMethod cash | gcash
String gcashRef GCash reference
String status approved | pending | voided
String payment_provider Embedded: gcash
String payment_intentId Embedded: UUID
String payment_status awaiting_payment | paid

3. Data Flow Diagram — Level 0 (Context Diagram)

High-level data exchange between external entities and the Mabuhay Homes Community Portal process boundary.



4. Data Flow Diagram — Level 1 (Full System Processes)

Decomposes Process 0 into 10 core operational sub-processes interacting with all 9 database stores.

